



DUBLIN CITY UNIVERSITY

SEMESTER ONE RESIT EXAMINATIONS 2013

MODULE: (Title & Code)	CA208 Logic
COURSE:	BSc. in Computer Applications (SE Stream)
YEAR:	2
EXAMINERS: (Including Telephone Nos.)	Dr. J. Power, Dr. M. Manzke, Dr. D. Sinclair, Ext no. 5510.
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Answer 10 Questions. Answer all 8 Questions in Section A. Answer any 2 questions from Section B. All questions carry equal marks.

Please do not turn over this page until instructed to do so

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones

Section A

QUESTION 1

[TOTAL MARKS: 10]

[10 Marks]

Describe a minimal set of operators for propositional logic. Show how the other operators can be expressed in terms of this minimal set of operators.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 10]

[10 Marks]

Describe, using an example, the relationship between *semantic tableaux* and *Gentzen System $\mathcal{G}$* .

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$\neg(p \vee q \rightarrow p \wedge q)$$

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 10]

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash ((A \rightarrow B) \rightarrow A) \rightarrow A$$

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 10]****[10 Marks]**

Using the Hilbert System $\mathcal{H}$ and the following derived rule

$$\text{Deduction: } \frac{U \cup \{A\} \vdash B}{U \vdash A \rightarrow B}$$

prove

$$\frac{\vdash \neg B \quad \vdash A \rightarrow B}{U \vdash \neg A}$$

[End of Question 5]

QUESTION 6**[TOTAL MARKS: 10]****[10 Marks]**

Using resolution prove the validity of

$$A \rightarrow (B \rightarrow (A \wedge B))$$

[End of Question 6]

QUESTION 7**[TOTAL MARKS: 10]****[10 Marks]**

Using Gentzen System $\mathcal{G}$ prove

$$\exists x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \exists x.q(x))$$

[End of Question 7]

QUESTION 8**[TOTAL MARKS: 10]****[10 Marks]**

Using Hilbert System $\mathcal{H}$ and the following derived rule and theorem

$$\text{Deduction: } \frac{U \cup \{A\} \vdash B}{U \vdash A \rightarrow B}$$

$$\text{Theorem: } p(a) \rightarrow \exists x.p(x)$$

prove

$$\vdash \forall x.p(x) \rightarrow \exists x.p(x)$$

[End of Question 8]

Section B

QUESTION 9

[TOTAL MARKS: 10]

[10 Marks]

Consider the following grid of points.

a	-	b	-	c	-	d
e	-	f	-	g	-	h
i	-	j	-	k	-	l
m	-	n	-	o	-	p

This grid can be represented by the following relations:

- `nexteast(X,Y)` is true if X is the immediate east point of Y, e.g. `nexteast(g,f)`.
- `nextsouth(X,Y)` is true if X is the immediate south point of Y, e.g. `nextsouth(m,i)`.

Write the following Prolog relations:

- `south(X,Y)` is true if X can be reached from Y by only travelling directly south.
- `east(X,Y)` is true if X can be reached from Y by only travelling directly east.
- `southeast(X,Y)` is true if X can be reached from Y by only travelling directly south or east.

[End of Question 9]

QUESTION 10

[TOTAL MARKS: 10]

[10 Marks]

Let a bus timetable be represented by a list of routes where each route is represented by `Start/End/DepartureTime/ArrivalTime`. Both `DepartureTime` and `ArrivalTime` are represented by a relation `time(Hour,Minutes)`. Write a Prolog relation `journey(Start, End)` that is true if there is a route in the timetable can get a passenger from `Start` to `End` directly or with one intermediate stop where there is at least 10 minutes between the arrival at an intermediate point before the departure to the destination of the journey. Assume the clock is a 24 hour clock and all departure and arrival times occur within the same day

[End of Question 10]

QUESTION 11**[TOTAL MARKS: 10]****[10 Marks]**

Let a set be represented by a list of elements. Write the following Prolog relations:

- `union(A,B,C)` is true if the list C contains only elements in either lists A or B.
- `intersection(A,B,C)` is true if the list C contains only elements in both lists A and B.

[End of Question 11]

Appendix

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \downarrow B_2)$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \downarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

Hilbert Logic

The *Hilbert System* $\mathcal{H}$ for propositional logic has 3 axioms,

Axiom 1: $\vdash (A \rightarrow (B \rightarrow A))$

Axiom 2: $\vdash (A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$

Axiom 3: $\vdash (\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$

and only 1 rule of inference.

Modus Ponens:
$$\frac{\vdash A \quad \vdash A \rightarrow B}{\vdash B}$$

The *Hilbert System* $\mathcal{H}$ for first-order logic has 2 additional axioms,

Axiom 4: $\vdash \forall x.A(x) \rightarrow A(a)$ x is a free variable in $A(x)$

Axiom 5: $\vdash \forall x.(A \rightarrow B(x)) \rightarrow (A \rightarrow \forall x.B(x))$ x is a free variable in $B(x)$
 x is not free in $A(x)$

and 1 additional rule of inference.

Generalisation:
$$\frac{\vdash A(a)}{\vdash \forall x.A(a)}$$

[END OF EXAM]